

EdumentX

Location-Based Tutor Discovery Platform
Design Improvement & Solution Specification

Comprehensive Analysis of 17 Design Discussion Points

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Executive Summary

This document is the result of a focused design discussion session on EdumentX, analyzing 17 critical areas for improvement. Each section presents the identified problem, a concrete solution design, implementation guidance, and UI/UX specifics. Two additional uncovered areas are flagged at the end. This document is intended as the authoritative design specification to guide both Figma prototyping and React Native development.

Point 1 Consistent Color Palette

Problem

Inconsistent or ad-hoc color choices across screens create a fragmented visual identity, reduce trust, and make the app feel unpolished rather than like a company-grade product.

Solution: Defined 4-Tier Color System

Use a fixed, named token system where every color in the app maps to a predefined role. No color is used outside this system.

Primary Brand — Education Blue

Main actions, headers, active navigation, primary buttons, links. Hex: #1A56DB (default), #0C3A7A (pressed/dark), #E8F0FE (light fill/backgrounds).

Trust Accent — Verification Teal

Blue Tick badge, verified tutor states, accepted enrollment, success toasts. Hex: #0D9E75 (active), #E6F7F2 (light fill).

AI Identity — Deep Purple

Everything related to the AI chatbot: chatbot header, AI message bubbles, AI suggestion chips. Hex: #4F46E5 (active), #EEF2FF (light fill). This color is RESERVED exclusively for AI — never use it elsewhere.

Semantic States (Amber/Red/Green/Gray)

Amber #B45309 for pending/warning. Red #DC2626 for error/rejected. Green #16A34A for completed/active. Gray #6B7280 for disabled/secondary. Each has a 50 (light fill) version for backgrounds.

Typography Palette

- Primary text: #111827 (near-black, never pure black #000000)
- Secondary text: #4B5563 (muted, for descriptions and metadata)
- Tertiary / placeholder: #9CA3AF
- On dark backgrounds (header bars, blue buttons): always #FFFFFF

Surface Palette

- Page background: #F9FAFB
- Card surface: #FFFFFF with 0.5px border #E5E7EB
- Elevated sheet (modals, bottom sheet): #FFFFFF with subtle top shadow line only
- Disabled state: #F3F4F6 fill with #9CA3AF text

Figma Setup Rule

Create a Color Styles library in Figma with every token named: brand/primary, brand/primary-light, trust/teal, trust/teal-light, ai/purple, ai/purple-light, state/pending, state/success, state/error, surface/page, surface/card, text/primary, text/secondary.

Every single design element must reference a style token, never a raw hex. This ensures global updates propagate automatically.

Point 2 Authentication System — Signup, Login & Admin Access

Problem

Using OTP for every login is slow and irritating for daily users. Admin having a completely separate portal adds development complexity. The current design has no clear separation between first-time signup and returning login.

Solution: Dual-Mode Auth with Password-Protected Login

Signup Flow (First Time Only)

1. User enters phone number on the Phone Entry screen.
2. System sends a 6-digit OTP via SMS. OTP screen shows 60-second countdown and a Resend option.
3. After OTP is verified, user is taken to Create Password screen. Password must be minimum 8 characters, one uppercase, one number. A confirm password field confirms no typos.
4. User is then taken to Role Selection screen (Student/Parent or Tutor). This screen is only shown ONCE and the choice is saved permanently to their profile.
5. User completes their role-specific profile setup (name, photo, subjects, location, etc.).
6. Account is created. User is logged in automatically and goes to their role-specific dashboard.

Login Flow (Every Subsequent Login)

7. Login screen shows: Phone number field + Password field. No OTP involved.
8. A 'Forgot Password?' link triggers a phone OTP reset flow.
9. After successful login, the system reads the stored role from the user record and routes directly to the correct dashboard — no role selection screen shown.

Admin Login — Same System, No Separate Portal

Admin accounts are just user records in the database with the role field set to 'admin'. The admin account is created once manually by the development team directly in Firestore (no public signup for admin). The same login screen is used. When the system detects role = 'admin' after login, it routes to the Admin Dashboard instead of a student or tutor view.

Security Implementation

Passwords are hashed with bcrypt (salt rounds: 12) before storage. Never store plain text.

Session tokens: JWT stored in device secure storage (not AsyncStorage). Token expiry: 7 days with silent refresh.

OTP is rate-limited: max 3 OTP sends per phone per hour to prevent SMS abuse.
Admin credentials must use a strong password (16+ chars) and are never accessible via any public-facing API.

Point 3 Role Selection — Signup Only

Solution

Role selection (Student/Parent vs Tutor) is shown exactly once: immediately after OTP verification during the signup flow. The selected role is stored in the user's Firestore document as a permanent field ('role': 'student' | 'tutor' | 'admin').

Login Behavior

- After login, the app reads the user's role from their stored profile.
- Student/Parent role → routes to Student Home screen.
- Tutor role → routes to Tutor Dashboard.
- Admin role → routes to Admin Panel.
- No role selection screen is ever shown again after signup.

Role Change Policy

A user cannot self-change their role after account creation. If a college student tutor graduates and wants to become a professional tutor, they submit a request through a 'Change Tutor Type' option in Profile Settings, which goes to admin review. A student/parent cannot switch to a tutor role without admin approval and re-verification. This prevents role abuse.

UI Note

The Role Selection screen must use two large, visually distinct cards with clear illustrations — a student studying (student/parent) and a person teaching on a whiteboard (tutor). The selection must feel definitive and clear since it cannot be changed later. A brief one-line description under each option explains what each role can do on the platform.

Point 4 Dashboard Design — Per Role

Student/Parent Dashboard

Layout Structure

10. Top bar: Greeting ('Good morning, Suhan'), current location badge (tappable to change), notification bell icon.
11. Search bar: Full-width, prominent, with a filter icon on the right.
12. Quick action chips: Map Search | AI Chatbot | Nearby Verified Tutors | Browse Batches.
13. Active Enrollments section: Horizontal scroll of enrollment cards showing tutor name, subject, next session day, and a 'View Details' arrow.
14. Nearby Tutors section: Horizontal scroll of tutor cards (avatar, name, Blue Tick if verified, distance badge, star rating, monthly rate).
15. Recommended for You section: Powered by the KNN algorithm, showing tutors matched to the student's grade/subject/location stored in profile.

Tutor Dashboard

Layout Structure

16. Top bar: Greeting, availability toggle (Available / Busy), notification bell.
17. Metric cards row (4 cards): Active students | Pending requests (with badge count) | Average rating | Total months taught.
18. Today's Schedule section: Time-ordered list of sessions for today. Shows student name, subject, time slot, and location pin.
19. Pending Requests section: Cards showing student name, grade, subject, requested plan, and distance from tutor. Accept / Reject / Counter buttons.
20. Capacity meter: Visual bar showing 'X of Y student slots filled'. Tutor sets Y during profile setup.
21. My Batches quick link: Shows count of active batches.

Admin Dashboard

Layout Structure

22. Verification Queue card: Count of tutors awaiting document review with a 'Review Now' button.
23. KPI metric cards: Total registered users | Verified tutors % | Active enrollments | Platform average rating.
24. Recent Activity feed: Latest enrollments, new registrations, verification actions.
25. Weekly trend chart: Enrollments over the past 7 days as a simple bar chart.
26. Subject demand chart: Horizontal bar chart of most-requested subjects.
27. Flagged reviews queue: Reviews reported by users as inappropriate, awaiting moderation.

Design Rule for All Dashboards

Metric cards use a consistent visual: muted 13px uppercase label on top, large 28px/500 number below, a small trend indicator (up/down arrow with percentage change). Cards are white with a 0.5px border, arranged in a 2x2 grid on mobile. The dashboard should feel like a daily briefing — actionable and scannable in under 10 seconds.

Point 5 Group Batch System — Creation by Tutor from Enrolled Students

Core Design Principle

Batches are always tutor-initiated and always formed from students who are ALREADY individually enrolled with that tutor. A student cannot directly request to join a batch — they can only enroll 1-to-1 and then be invited by the tutor into a batch.

Batch Creation Flow

28. Tutor navigates to the Batches tab in the bottom navigation.
29. Taps 'Create New Batch' button.
30. A multi-select screen appears showing all currently enrolled students as cards. Each card shows: student photo, name, grade, subject, and current 1-to-1 schedule. Tutor selects 2–6 students who share the same subject and approximate schedule.
31. Tutor fills in batch details: batch name (e.g., 'Grade 10 Physics Group A'), subject, new shared monthly rate per student, proposed schedule (days of week + time slot).
32. Tutor taps 'Send Batch Invitation'. Each selected student receives a push notification: 'Your tutor [Name] has invited you to join a group batch for [Subject]. New rate: Rs X/month. Tap to view details.'
33. Students see a Batch Invitation card in their Enrollments tab. They can Accept or Decline the invitation.
34. If a student accepts, their 1-to-1 enrollment is replaced by the batch enrollment at the new rate. If they decline, their 1-to-1 enrollment continues unchanged.
35. The batch is activated once at least 2 students accept. If fewer than 2 accept after 48 hours, the batch invitation expires and all students remain on 1-to-1.

UI — Student Card Selection Screen

Display enrolled student cards in a 2-column grid. Each card is tappable to select (shows a blue checkmark overlay when selected). The header shows 'Select students for batch (2-6 students)' with a live counter. Subject filter chips at the top help the tutor quickly filter by subject when they have many enrolled students.

Point 6 Tutor View of Student Approximate Location

Problem

A tutor needs to know whether a student is within their service radius before accepting an enrollment request. However, showing the exact home address of a student on the map is a serious privacy violation.

Solution: Approximate Radius Disclosure

When a student sends an enrollment request, the system stores their precise GPS coordinates. However, on the tutor's Enrollment Request screen, only an approximate location is shown using a neighborhood-level radius circle, not the exact point.

Implementation Details

- The student's exact coordinates are obfuscated by adding a random ± 300 – 500 meter offset before displaying to the tutor. The tutor sees a radius circle on the map, not a pin.
- The request card shows: 'Student is approximately 2.1 km from you' (calculated using the real Haversine distance, shown as a rounded distance, not a map point).
- On the request detail screen, a small embedded map shows: (1) the tutor's location as a blue pin, (2) a teal shaded circle representing the approximate student area, (3) the tutor's service radius overlay in a lighter shade.
- If the student's approximate area overlaps with the tutor's service radius, a green banner shows: 'This student is within your service area.' If not, an amber banner shows: 'This student is slightly outside your set radius — consider if you can accommodate.'
- The exact address the student specified for teaching location is only revealed AFTER the tutor accepts the enrollment request, not before.

Privacy & Legal Note

No exact student address, pin, or coordinates are ever shown to a tutor before acceptance.

This is consistent with the project's legal feasibility section which states 'exact locations of tutors or students are not displayed publicly'.

After acceptance, the teaching address is shared in the enrollment details screen as it is operationally necessary for the tutor to conduct the session.

Point 7 Batch Fee Management & Rate Negotiation

Problem

When a tutor groups students into a batch, parents naturally expect a lower fee since the tutor is teaching multiple students simultaneously. This creates a negotiation situation that must be handled fairly — preventing rate disputes and protecting both tutor income and student value.

Solution: Transparent Tiered Rate Structure

The platform enforces a transparent rate structure where every tutor must set THREE rate fields on their profile during setup:

36. 1-to-1 Rate: Monthly rate for private individual sessions (the default).
37. Minimum Batch Rate: The lowest rate per student the tutor is willing to accept in a group setting. The system will not allow a batch rate below this value.
38. Recommended Batch Rate: A suggested per-student batch rate that auto-populates when creating a batch (tutor can adjust upward or downward to the minimum).

Batch Rate Display to Students

When a student views a tutor's profile, both rates are shown side by side in the 'Rate' section: '1-to-1: Rs 3,500/month | Group batch: from Rs 2,200/month per student'. This sets expectations transparently before any enrollment request is made.

Batch Invitation Rate Logic

- When the tutor creates a batch and sets the per-student rate, the invitation card shown to students must clearly display: the batch rate, the savings vs. their current 1-to-1 rate, and the number of other students in the batch.
- Students can Accept (at the tutor's stated batch rate) or Decline (staying at 1-to-1 rate). There is NO counter-offer from the student side — the rate negotiation happens before batches are formed, not after.
- Tutors cannot set a batch rate higher than their 1-to-1 rate. The system validates this and shows an error if attempted.

Anti-Abuse Rule

The system should log and flag any tutor who repeatedly creates and dissolves batch invitations within short periods, as this could be used to artificially lower rates without genuine batch formation. Admin can investigate flagged patterns.

Point 8 Enrollment Default — 1-to-1 Only

Solution

The enrollment request form submitted by a student is always and only a 1-to-1 private tuition request. There is no batch option in the enrollment request form. Students CANNOT request a batch directly.

Separate Batch Browsing Path

Students who want group learning have a separate path: the 'Browse Batches' section on the home feed and map screen. Here, tutors can publicly list open batch slots (seats available in an existing batch) that any student can browse and request to join. This is distinct from the main enrollment request flow.

Two Distinct Enrollment Entry Points

Standard Enrollment Flow	Batch Join Flow
Source	Source
Student sends enrollment request	1-to-1 enrollment form
Student requests to join listed open batch	Browse Batches → Request Join

- Standard Enrollment → Tutor receives in inbox as a 1-to-1 request → Accepts/Rejects/Counter-offers → Active 1-to-1 enrollment.
- Batch Join Request → Tutor receives a separate 'Batch Join Request' notification → Approves → Student directly joins the open batch (no 1-to-1 phase).
- Tutor can ALSO invite existing 1-to-1 students into a newly created batch (as described in Point 5).

Point 9 Time Shift & Scheduling System

Problem

The current design lacks a structured approach to session scheduling. Without a proper time-slot system, double-booking is possible, schedule communication is vague, and both tutors and students have no reliable way to track upcoming sessions.

Solution: Availability Calendar + Slot-Based Booking

Tutor Availability Setup

39. In the Tutor Profile / Dashboard, a 'Set Availability' section presents a weekly calendar grid.
40. Tutor taps each day of the week (Mon–Sun) and for each active day, selects time slots from a list: Morning 6am–9am, Morning 9am–12pm, Afternoon 12pm–3pm, Afternoon 3pm–5pm, Evening 5pm–7pm, Evening 7pm–9pm. Each slot is a 2-hour block.
41. Tutor can mark specific dates as Unavailable (holidays, exams, travel) using a date-range blocker.
42. All availability is stored in Firestore and is live — students only see slots the tutor has marked available.

Student Slot Selection During Enrollment

43. When a student fills the enrollment request form, after selecting subject and plan duration, they see a 'Select Schedule' section showing only the tutor's marked-available days and slots.
44. Student selects preferred days per week (e.g., Monday and Thursday) and preferred time slot (e.g., Evening 5pm–7pm).
45. The system checks for conflicts: if the tutor already has 2 students at that exact slot, it shows a 'Slot is full' indicator and asks the student to choose another time.
46. The selected schedule is stored in the enrollment record and shown to both parties.

Conflict Detection Rules

- A tutor can have at most 1 student per slot per session type (1-to-1). Batch students share one slot.
- If a tutor creates a batch, the batch occupies a full slot. No individual 1-to-1 can be assigned to that same slot.
- The system highlights already-booked slots in gray and available slots in teal on the selection calendar.

Session Reminders

Push notification to both tutor and student 60 minutes before each scheduled session.

A 'Sessions Today' card on the dashboard shows the day's upcoming sessions with time, student/tutor name, and location.

Session cancellation: either party can cancel with minimum 6-hour notice. Repeated cancellations by either party are tracked and visible on their profile.

Point 10 Responsive Design for All Mobile Screen Sizes

Target Devices

Screen Width	Example Device
Small Android (360px wide)	Samsung Galaxy A series, older phones
Standard iPhone (390px)	iPhone 14, iPhone 13
Large Android / iPhone Pro Max (430px+)	iPhone 14 Pro Max, Samsung S series
Tablet (600px+)	iPad, Samsung Tab (secondary support)

React Native Responsive Implementation

- Use `Dimensions.get('window').width` to get device width at runtime. Store in a `useWindowDimensions` hook.
- Font sizes: base 14px for body text on small screens, 15px on standard, 16px on large. Scale using a font scale multiplier ($\text{width} / 390$).
- Card padding: 12px on small, 16px on standard, 20px on large screens.
- Bottom navigation icon size: 22px on small screens, 26px on standard, 28px on large.
- Map pins: 32px on standard, 36px on large. Must never be smaller than 28px for tapability.

Safe Area Handling

- Always wrap screens in `SafeAreaView` from `react-native-safe-area-context`.
- Account for notches (iPhone Dynamic Island, punch-hole cameras) using `useSafeAreaInsets`.
- Bottom navigation bar height: device safe area bottom inset + 56px base height.
- Status bar content color: light (white icons) for dark header screens, dark (dark icons) for light content screens.

Figma Screen Setup

- Design all screens on iPhone 14 frame (390×844px) as the primary reference.
- Create one additional version of the Home, Map, and Dashboard screens at 360px width to verify no content clipping.
- Use Auto Layout throughout — every component must resize naturally. No fixed absolute positioning for content.
- Test all text that can be long (tutor names, subject lists) with truncation ellipsis after 1 or 2 lines maximum.

Point 11 Map View Improvements

Current Limitations

A basic pin-drop map with no clustering, no visual distinction between tutor types, and no interactive preview loses most of its value to users who need to quickly compare nearby options at a glance.

Improved Map Feature Set

Pin Clustering

- When zoomed out (zoom level < 14), nearby pins are grouped into a cluster circle showing the count (e.g., a blue circle with '12'). Tapping a cluster zooms into that area.
- Prevents visual clutter in dense areas like Kathmandu core where many tutors may be concentrated.

Differentiated Pin Styles

- Standard tutor (unverified): solid blue circle with a person silhouette icon.
- Verified tutor (Blue Tick): solid teal circle with a checkmark-person icon. Slightly larger (36px vs 28px) to stand out.
- At-capacity tutor: gray circle with a lock icon overlay. Still visible but cannot receive enrollment requests.
- Selected / tapped pin: enlarges to 44px, shows a white border ring around it.

Service Radius Overlay

- When a tutor pin is tapped, a semi-transparent teal circle (20% opacity fill, 1px solid border) appears showing that tutor's service radius. This helps students immediately see if they are within range.
- The radius overlay disappears when another pin is tapped or the map is panned away.

Floating Mini Info Card

- Tapping a tutor pin shows a compact info card sliding up from the bottom (200px height) WITHOUT navigating away from the map. Card shows: tutor photo, name, Blue Tick if verified, subject chips, distance, rating, monthly rate, and two buttons: 'View Profile' and 'Quick Enroll'.
- This allows students to evaluate a tutor without losing their map context.

Map Controls

- 'Near me' button (GPS arrow icon): re-centers the map to the student's current location.
- Radius slider: floating pill at 28% from top showing 'Showing tutors within X km' — adjustable from 1 to 20 km. Pins update in real time as the slider moves.
- Layer toggle: switch between 'All tutors' and 'Verified only' filter directly on the map.
- Street view is intentionally excluded — this is a tutor discovery tool, not a navigation app.

Point 12 Verification System — Blue Tick, Phone Uniqueness & Admin Review

Phone Number as Identity Anchor

Every account on EdumentX is anchored to a unique phone number. The database enforces that no two accounts can share the same phone number. Since phone numbers in Nepal require NTC/Ncell SIM registration with a citizenship ID, this creates a real-world identity link without the platform needing to re-verify citizenship for basic accounts.

Three-Tier Verification System

Tier 0 — Phone Verified (All Users)

Automatically granted upon successful OTP signup. Shown as a small gray checkmark on profiles. Means: 'This is a real phone-verified account.' Applies to all users including students.

Tier 1 — Student Tutor Badge (College Students)

Requires: Upload of valid college/university student ID showing current enrollment. Upload of citizenship ID or NID. Profile must be at least 30 days old. Awarded by admin after document review. Badge shown as a blue shield with a graduation cap icon.

Tier 2 — Blue Tick — Verified Professional

Requires: Citizenship ID, academic degree/teaching certificate, minimum 2 years of verifiable experience, a 1-minute teaching demo video reviewed by admin, profile fully complete with all subjects and availability filled. Badge: the teal Blue Tick. This is the highest trust signal on the platform — visible prominently on the tutor card and profile.

Admin Verification Workflow

47. Tutor uploads required documents and video. Their profile shows 'Verification Pending' amber banner.
48. Admin opens the Verification Queue. Each queue item shows: tutor name, submission date, document thumbnails with expand-to-fullscreen capability, video player for the demo video.
49. Admin can: Approve (grants the appropriate badge), Reject with reason (tutor gets a notification explaining what was missing), or Request More Info (sends a message requesting a specific document resubmission).
50. Approved tutors receive a push notification: 'Congratulations! Your EdumentX Verification is complete. Your [badge type] badge is now live on your profile.'
51. If a tutor's documents are found to be fraudulent at any time, admin can Revoke badge and Suspend account.

Document Privacy & Storage

All uploaded documents are stored in Firebase Storage under a private bucket — inaccessible via public URL.

Only authenticated admin accounts can generate time-limited signed URLs to view documents.

Documents are automatically deleted 90 days after account deletion.

The demo video is reviewed by admin but never shown publicly to students — only the badge result is shown.

Point 13 Bottom Navigation Layout — Per Role

Design Rule

The bottom navigation is the primary navigation structure of the app. Each role gets exactly 4 or 5 icons (no more — beyond 5 creates cognitive overload on mobile). All navigation items are always visible, never hidden in a hamburger menu. Settings are accessed through the Profile tab, not as a standalone bottom nav item.

Student/Parent Bottom Navigation (5 tabs)

Tab	Icon
Home	House icon
Map	Map pin icon
AI Chat	Sparkle/wand icon (purple)
Enrollments	Calendar icon
Profile	Person circle icon

- Active tab: icon fills with brand blue, label appears below in blue.
- Inactive tabs: outline icon, no label (label only on active to save space).
- AI Chat tab uses purple (the AI identity color) even when inactive — this makes it instantly identifiable as the AI feature.
- Notification badge (red dot with count) appears on the Enrollments tab when there are pending batch invitations or enrollment status updates.

Tutor Bottom Navigation (4 tabs)

Tab	Icon
Dashboard	Grid/home icon
Inbox	Message-inbox icon
Batches	People-group icon
Profile	Person circle icon

- Inbox tab shows a red badge count for unread enrollment requests.
- Batches tab shows a badge for pending batch join requests.
- Profile tab contains: Edit Profile, My Documents, Availability Settings, Verification Status, App Settings, Log Out.

Admin Bottom Navigation (4 tabs)

Tab	Icon
Queue	Shield-check icon
Users	Users icon
Stats	Chart-bar icon
Settings	Cog icon

Implementation Note

Use React Navigation's Tab Navigator with a custom tab bar component. Each tab icon should be from a consistent icon library — react-native-vector-icons (MaterialCommunityIcons) is recommended for consistent weight and style. Icon size: 26px standard on all devices, scaled using the responsive multiplier from Point 10.

Point 14 College Student Tutors vs Professional Tutors

Why This Distinction Matters

The primary target audience (college students doing part-time tutoring) differs from professional/retired teachers in several important ways: verification requirements, credibility, price range, and the subjects they can teach. The platform must handle both without creating a confusing or unfair experience for either group.

Tutor Type Selection During Signup

After selecting the Tutor role during signup, the user is asked a secondary question: 'What best describes you?' with two options:

- Student Tutor: Currently enrolled in a college or university, tutoring part-time.
- Professional Tutor/Teacher: A qualified teacher, retired educator, or full-time private tutor.

This choice determines the verification pathway, the displayed badge, the recommended price bracket, and how the RAG chatbot and KNN algorithm weigh their profiles in recommendations.

Verification — Student Tutor Path

- Required documents: Valid college enrollment certificate OR student ID card (current academic year), citizenship ID/NID.
- Badge awarded: Blue cap-and-book icon (Student Tutor badge).
- The badge communicates 'verified college student' to parents. It is not the Blue Tick (which is reserved for professional-level verification).
- Student tutors are reminded to update their enrollment certificate annually.

Verification — Professional Tutor Path

- Required documents: Citizenship ID, degree certificate (bachelor's or above in the subject area), teaching license if applicable, 1-minute demo video.
- Badge awarded: The Blue Tick (Tier 2 in the system from Point 12).
- Professional tutors have a separate 'Experience' field where they enter years of teaching, previous institutions, and can optionally add reference contacts.

Impact on RAG Chatbot & KNN Recommendations

- When a parent asks the AI chatbot 'Find a trustworthy experienced tutor for Grade 10 Maths', the system should prioritize Blue Tick professional tutors.
- When a college student asks 'Find an affordable tutor for Physics who understands university-level content', the system should give weight to similarly-aged student tutors at lower price points.
- The RAG context injected into the AI prompt must include tutor type, verification tier, and years of experience as filterable fields.

- The KNN feature vector (from the original proposal) should include a 'verification tier' feature with weight 0.10, redistributed from the 'experience' weight.

Point 15 Platform Quality — Making EdumentX a Proper Education Platform

Beyond Discovery: Platform Features That Build Retention

A discovery-only app risks being used once (when a student finds a tutor) and then abandoned. The following features transform EdumentX from a finder tool into an ongoing platform where both tutors and students return regularly.

For Tutors

- Profile completion score: A progress bar (e.g., 'Your profile is 70% complete') with specific prompts — Add a demo video | Fill in all subjects | Add availability. A complete profile is weighted higher in search results.
- Response rate tracking: The system tracks how quickly tutors respond to enrollment requests. Tutors with >90% response rate within 24 hours get a 'Responsive' badge on their profile.
- Monthly Top Tutor recognition: The highest-rated, most-enrolled verified tutor in each subject area receives a 'Top Tutor of the Month' badge. Displayed on the tutor card for 30 days. Encourages quality competition.
- Analytics for tutors: In the Profile tab, tutors can see a simple analytics screen showing profile views per week, enrollment request rate, and average rating trend over time.

For Students & Parents

- Saved tutors list: Students can bookmark tutor profiles with a heart/save icon. Saved tutors appear in a 'My Saved Tutors' section in the Profile tab.
- Enrollment history: Complete record of all past enrollments with tutor name, subject, duration, dates, and whether a review was left.
- Re-enroll shortcut: On past enrollment cards, a 'Re-enroll with [name]' button appears, pre-filling the enrollment form with the previous details.
- Tutor recommendations after completion: When an enrollment ends, the app suggests 3 similar tutors based on subject and rating — encouraging continued use of the platform.

Platform-Wide

- Cancellation rate: Both tutors and students have a visible cancellation rate metric on their profile, so both parties can assess reliability before committing.
- Dispute resolution: A 'Report Issue' option in active enrollments allows either party to flag problems (no-shows, payment disputes, inappropriate behavior) for admin review.
- Community guidelines displayed during signup — clear rules about conduct, honesty of credentials, and rate transparency.

Point 16 Tutor Capacity Limits — Preventing Overloading

Problem

Without a capacity limit enforced by the platform, a motivated (or greedy) tutor could accept unlimited students and batches, resulting in poor teaching quality for all students. This damages the platform's reputation and student outcomes.

Solution: System-Enforced Capacity Cap

Mandatory Capacity Setting

During profile setup, every tutor **MUST** set their maximum student capacity. This field cannot be left empty. The system presents guidance based on tutor type:

Tutor Type	Capacity Guidance
Student Tutor	Recommended: 3–6 students total. System maximum allowed: 10 students.
Professional Tutor	Recommended: 5–12 students total. System maximum allowed: 20 students.

How Capacity is Counted

- Each individual 1-to-1 enrollment counts as 1 toward the tutor's capacity.
- A batch of 4 students counts as 4 toward capacity (because the tutor is managing 4 real student relationships).
- Total capacity = sum of all individual enrollments + sum of all students in all active batches.

System Enforcement

- When a tutor is at capacity, their profile map pin changes to the gray 'at-capacity' lock icon. A banner on their profile reads: 'This tutor is currently at full capacity. You can join their waitlist.'
- The enrollment form 'Submit' button is disabled for that tutor with a tooltip: 'Tutor is at full capacity.'
- The tutor's new enrollment requests are blocked server-side in Firebase Cloud Functions as a secondary enforcement — client-side blocking alone is insufficient.

Waitlist System

- Students can join a tutor's waitlist (limited to 5 waitlist slots per tutor).
- When a student leaves (enrollment ends or is cancelled), the tutor is notified and the first waitlisted student is automatically notified: '[Tutor Name] now has an open slot — enroll before it fills up.'

Anti-Gaming Rule

Tutors cannot set their capacity to 1 to game the waitlist system and create artificial scarcity.

Minimum capacity: 2 students.

Admin can audit tutors whose capacity changes frequently or who have unusually high cancellation/re-acceptance patterns.

Point 17 Improved Rating & Review System

Problem

A single 5-star rating is too simplistic to give parents meaningful information about a tutor. It does not reveal what the tutor is specifically good or bad at, does not prevent fake reviews, and does not give tutors actionable feedback to improve.

Solution: Multi-Dimensional Verified Review System

Review Dimensions (5 Categories)

Dimension	What it measures
Teaching Quality	Is the teaching clear and effective?
Punctuality	Does the tutor arrive on time?
Communication	Is the tutor responsive and approachable?
Subject Knowledge	Does the tutor know the subject deeply?
Overall Experience	Would you recommend this tutor?

Each dimension is rated 1–5 stars independently. The 'Overall Experience' star rating is the one shown prominently on the tutor card. The other four dimensions are shown in a breakdown panel on the tutor profile page, visible on expanding a 'See rating breakdown' section.

Eligibility Rules — Preventing Fake Reviews

- Only users with an active OR completed enrollment with that specific tutor can leave a review.
- Reviews can only be submitted after a minimum of 2 confirmed sessions have been completed (tracked via the scheduling system from Point 9).
- Each enrollment can generate exactly 1 review — no duplicate reviews per enrollment.
- Reviews cannot be edited after submission (to prevent post-dispute manipulation), but the student can flag their own review for deletion by contacting admin.

Review Content

- Written review: Required, minimum 30 characters, maximum 500 characters.
- Auto-generated completion tags: After submitting, students can optionally select tags from a set: 'Very clear explanations', 'Patient with questions', 'Always on time', 'Great for exam prep', 'Affordable', 'Good for beginners'. Tags appear as chips on the review.
- Photo of session materials (optional): Student can attach a photo of notes/work done in the session. This adds authenticity.

Tutor Response

- Tutors can write a single public response (max 200 characters) to each review. Response appears below the review.
- Tutors cannot delete reviews, but can flag reviews they believe are fraudulent or violate guidelines.

Rating Display on Profile

- Overall star rating: Large display (e.g., 4.7 ★) with total review count in parentheses.
- Star distribution: Horizontal bar chart showing how many 5/4/3/2/1 star reviews exist.
- Dimension breakdown: Five small bars showing average score for each of the 5 dimensions.
- Recent reviews shown first. Toggle to 'Highest rated' or 'Most critical' sort order.

Moderation

Admin review queue for flagged reviews. Admin can delete reviews that violate guidelines.

Reviews containing personal information, profanity, or unverifiable claims about tutor behavior outside sessions are subject to removal.

A 'Verified Enrollment' badge appears on each review, confirming the reviewer genuinely enrolled with the tutor — this is the key trust signal for parents.

Additional Areas Not Yet Discussed

The following two areas were not explicitly covered in the 17 discussion points but are critical for a complete, production-ready app. They should be addressed before final development begins.

Point A Notification System Design

The app relies heavily on real-time communication between students, tutors, and admin. A well-designed notification architecture is essential.

Notification Types Needed

- Enrollment request received (tutor)
- Enrollment accepted / rejected / counter-offered (student)
- Batch invitation received (student)
- Session reminder — 60 minutes before (both parties)
- Waitlist slot opened (waitlisted student)
- Verification status update (tutor)
- New review received (tutor)
- Admin message about document resubmission (tutor)

Implementation

- Firebase Cloud Messaging (FCM) for push notifications — already in the tech stack.
- In-app notification center (bell icon in top bar) showing the last 30 notifications with read/unread state.
- All notifications are stored in Firestore under the user's notifications sub-collection for retrieval.
- Notification preferences screen in Profile Settings: students and tutors can toggle off specific notification types.

Point B Offline & Low Connectivity Handling

Kathmandu Valley has good internet coverage in most areas, but mobile data can be intermittent in some neighborhoods. The app should handle this gracefully.

- Tutor profiles and search results should be cached locally (using React Query or Firestore offline persistence) so the last-loaded data is visible even without internet.
- Form submissions (enrollment requests, reviews) should queue locally when offline and auto-submit when connectivity returns. A small banner at the top of the screen shows 'You are offline — actions will sync when connected.'
- The map view degrades gracefully: if Google Maps tiles fail to load, a placeholder message shows 'Map unavailable — switch to list view' and the list view with distance data (calculated from cached coordinates) is shown automatically.

- No critical data (user authentication tokens, current enrollment details) should rely solely on network requests for display — always cache the most recent state locally.

Implementation Priority Order

Based on the analysis above, the following sequence is recommended for the 14-week sprint timeline:

Sprint	Focus Areas
Sprint 1–2	Auth system (Point 2, 3), Color system tokens (Point 1), Bottom navigation (Point 13)
Sprint 3–4	Student/Tutor profile setup, Verification upload (Point 12, 14)
Sprint 5–6	Map search + pin system (Point 11), Schedule/availability (Point 9)
Sprint 7–8	Enrollment flow 1-to-1 (Point 8), Tutor request view with location (Point 6)
Sprint 9–10	Batch system — tutor creation from students (Point 5), Batch fees (Point 7)
Sprint 11–12	Dashboard improvements (Point 4), Rating system (Point 17), Capacity limits (Point 16)
Sprint 13–14	AI chatbot integration, Notifications (Point A), Responsive testing (Point 10), Offline handling (Point B)

This document covers all 17 discussion points plus 2 additional critical areas (Notifications and Offline Handling). Every solution is designed to be feasible within the free-tier Firebase stack and React Native (Expo) technology choices outlined in the original project proposal.